Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2026

Course: B.Tech | *Time: 03 Hours*

Code: 100602 | *Subject: Computer Networks* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.

*(ii) There are **NINE** questions in this paper.*

*(iii) Attempt **FIVE** questions in all.*

(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): [2 x 7 = 14]

(a) A fully connected mesh network contains 8 nodes. Every pair of nodes requires a dedicated point-to-point link. How many links are required?

(i) 16 (iii) 32

(ii) 28 (iv) 56

(b) Wavelength Division Multiplexing is primarily associated with

(i) Copper cables (iii) Optical fiber communication

(ii) Radio communication only (iv) Coaxial cable only

(c) A communication code has minimum Hamming distance $d_{min} = 5$. What is the maximum number of errors that can be detected?

(i) 2 (iii) 5

(ii) 3 (iv) 4

(d) A sender transmits frame 5 but does not receive its acknowledgement within the timeout interval. Assume the frame or its acknowledgement has been lost. What should the sender do?

(i) Retransmit frame 5 (iii) Discard all previously transmitted frames

(ii) Move directly to frame 6 (iv) Restart the entire connection

(e) Which statement correctly distinguishes IPv6 from IPv4?

(i) IPv6 uses 32-bit addresses, while IPv4 uses 128-bit addresses (iii) IPv6 uses 128-bit addresses, while IPv4 uses 32-bit addresses

(ii) Both use 32-bit addresses (iv) Both use 128-bit addresses

(f) Host A knows the IP address of Host B, and both hosts are on the same LAN. Host A does not have Host B's MAC address in its ARP cache. What happens first?

(i) Host A broadcasts an ARP request asking for the MAC address corresponding to Host B's IP address. (iii) Host B broadcasts its MAC address.

(ii) Host A unicasts the IP packet directly to Host B. (iv) The router assigns a new IP address to Host A.

(g) An application requires reliable, ordered delivery with retransmission of lost data. Which transport protocol is more appropriate?

(i) UDP (iii) ARP

(ii) TCP (iv) DHCP

(h) A client wants to access www.example.com but does not know its IP address. Which application-layer service is responsible for resolving the domain name?

(i) FTP (iii) TELNET

(ii) DNS (iv) SNMP

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(i) Consider the following TCP exchange:
(i) Client → Server: SYN
(ii) Server → Client: SYN + ACK
(iii) Client → Server: ACK
What is the purpose of this exchange?
(i) IP fragmentation
(ii) TCP connection termination
(iii) UDP connection establishment
(iv) TCP connection establishment

(j) A web browser sends a request to a web server and receives a response containing the requested resource. Which communication model is primarily illustrated?
(i) Request-response
(ii) Token passing
(iii) Broadcast
(iv) Error correction

Q.2 (a) Explain the layered architecture of the OSI Reference Model with a neat block diagram. Describe the primary functions, data units and responsibilities of each of the seven layers. [7]

(b) Explain Mesh and Star network topologies with suitable diagrams. Compare their advantages and disadvantages. [7]

Q.3 (a) Explain unguided transmission media used in computer networks with suitable examples. Discuss their characteristics, advantages and disadvantages. [7]

(b) Explain wired LAN and wireless LAN. Compare their characteristics, advantages and limitations. [7]

Q.4 (a) Explain Time Division Multiplexing with a suitable diagram. Discuss its working and advantages. [7]

(b) Compare Frequency Division Multiplexing, Time Division Multiplexing and Wavelength Division Multiplexing based on signal type, medium, operation and applications. [7]

Q.5 (a) Explain the principle of error detection using Cyclic Redundancy Check (CRC). Given a dataword 1101011011 and a generator polynomial $G(x) = x^4 + x + 1$, determine the transmitted codeword using binary modulo-2 division. Verify how the receiver detects errors if no bit is altered. [7]

(b) Explain the Stop-and-Wait ARQ protocol with a suitable diagram. Discuss its working. [7]

Q.6 (a) Describe the IPv4 header format with a suitable diagram. Explain the functions of the TTL, Header Checksum, Flags and Fragmentation Offset fields. [7]

(b) Explain the working of ARP in mapping an IP address to a MAC address. Describe the ARP request and ARP reply process with a neat diagram. [7]

Q.7 (a) Describe the connection establishment and connection termination mechanisms in TCP with sequence timing diagrams. Explain the role of SYN, ACK and FIN control flags and explain the significance of the TIME_WAIT state during connection teardown. [7]

(b) Explain the Leaky Bucket algorithm with a suitable diagram. Discuss its working and applications. [7]

Q.8 (a) Explain the hierarchical architecture of the Domain Name Space. Describe the domain name resolution process using recursive and iterative queries with a suitable sequence diagram. [7]

(b) Describe the working of File Transfer Protocol with a neat diagram. Describe the control connection and data connection used for file transfer. [7]

Q.9 Write short note on ***any two*** of the following: [7 x 2 = 14]
(a) Virtual LAN
(b) Piggybacking
(c) Quality of Service
(d) Bluetooth

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